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Community aware temporal network generation

Nicolò Alessandro Girardini[1,2*], Antonio Longa[2], Gaia Trebucchi[2], Giulia Cencetti[3], Andrea Passerini[2] and Bruno Lepri[1]

*Correspondence:
Nicolò Alessandro Girardini
ngirardini@fbk.eu;
nicolo.girardini@unitn.it
[1]Mobile and Social Computing Lab (MobS), Fondazione Bruno Kessler (FBK), Trento, Italy
[2]Department of Information Engineering and Computer Science (DISI), University of Trento, Trento, Italy
[3]Université de Toulon, CNRS, CPT, Aix-Marseille Université, Marseille, France

Abstract
The advantages of temporal networks in capturing complex dynamics, such as diffusion and contagion, has led to breakthroughs in real world systems across numerous fields. In the case of human behavior, face-to-face interaction networks enable us to understand the dynamics of how communities emerge and evolve in time through the interactions, which is crucial in fields like epidemics, sociological studies and urban science. However, state-of-the-art datasets suffer from a number of drawbacks, such as short time-span for data collection and a small number of participants. Moreover, concerns arise for the participants' privacy and the data collection costs. Over the past years, many successful algorithms for static networks generation have been proposed, but they often do not tackle the social structure of interactions or their temporal aspect. In this work, we extend a recent network generation approach to capture the evolution of interactions between different communities. Our method labels nodes based on their community affiliation and constructs surrogate networks that reflect the interactions of the original temporal networks between nodes with different labels. This enables the generation of synthetic networks that replicate realistic behaviors. We validate our approach by comparing structural measures between the original and generated networks across multiple face-to-face interaction datasets.

Keywords Temporal networks, Face to face interactions, Communities

Introduction

The rise of network science in recent times fostered a great amount of innovation in multiple and diverse research fields (Newman 2010; Barabasi 2016), from classical ones, such as biology (Arrell and Terzic 2010; Kovács et al. 2019), psychology (Beard et al. 2016; Borsboom et al. 2021), economics (Jackson 2008), to newer ones, such as urban science (Batty 2013; Alessandretti et al. 2020; Mauro et al. 2022; Bontorin et al. 2024) and neuroscience (Vico Fallani et al. 2014; Bassett and Sporns 2017). Moreover, network science naturally provides tools to understand how humans behave and interact among themselves on many levels, e.g., temporal level and group (or community) level (Lazer et al. 2009; Peixoto and Rosvall 2017; Gelardi et al. 2021; Galdeman et al. 2023; Dall'Amico et al. 2024). In fact, it is easy to associate nodes with individuals and links between them with an interaction that occurs, or some other possible tie, such as being relatives.

Studying human interactions, specifically face-to-face or proximity interactions, became particularly relevant with the use of sensors that could easily be worn or with the use of smartphones. In this way, study participants could carry individual devices during their daily activities, allowing the devices to record their behavior in a natural setting. These studies are defined *Living Labs* (Eagle and Pentland 2006; Cattuto et al. 2010; Aharony et al. 2011; Isella et al. 2011; Stopczynski et al. 2014; Centellegher et al. 2016; Nepal et al. 2022) and proved to be very useful to study human interactions and their behavioral routines (Eagle and Pentland 2009; Starnini et al. 2017; Ozella et al. 2019; Iacopini et al. 2022; Girardini et al. 2023; Lotito et al. 2024; Arregui-García et al. 2024). Nevertheless, the problems of recruitment and keeping the engagement high are significant for this kind of data collection, and resulting data are often limited in population size and temporal length. These limitations affect the possibility of using the corresponding temporal networks for simulating realistic dynamics, such as epidemic spreading or opinion dynamics.

A possible solution to this problem is represented by the use of synthetic networks, that are generated to reproduce specific characteristics of real networks (Pereira et al. 2020; Zeno et al. 2021; Zeno and Neville 2023; Cüppers et al. 2024). Among the synthetic networks that can be built, surrogate networks, i.e., synthetic networks generated to imitate a specific input network, are particularly relevant in this task. In fact, they can be used as suitable replacements to the original real network and, if the generation model allows it, the surrogate networks are not bound to have the same temporal length and population size of the networks they are imitating (Laurent et al. 2015; Purohit et al. 2018; Cencetti and Barrat 2024). The model we propose possesses these characteristics, enabling the generation of realistic synthetic datasets without constraints on temporal length or network size.

We build our methodology upon a pre-existing method, namely *ETN-gen* (Longa et al. 2024), which generates surrogates that reproduce the local behavior of nodes. The idea of this generation process is to first decompose the original network into small temporal subnetworks, the Egocentric Temporal Neighborhoods (ETN), that encode the local neighborhood of each node on a short time-scale. The ETN can then be used as building blocks to build the surrogate networks. A limitation of the *ETN-gen* method is that it does not account for the heterogeneity of nodes. As a result, the behavior of all nodes in the generated network, while stochastically distinct, is derived from the same underlying patterns shared by all nodes. However, real-world networks often exhibit a modular structure. In social environments, the interplay between individuals with different roles or characteristics was shown to play a crucial role in shaping their distinct behaviors and interactions (Guimera et al. 2003; Hoffman et al. 2020). This is true for the temporal component as well, as frequency of interactions and duration change according to the described interplay (Stopczynski et al. 2015; Sekara et al. 2016). An example is the Reality Mining behavioral study (Eagle and Pentland 2009), which showed different interaction patterns for economics students, first year students and senior students. Other environments also share these differences. For example, in a hospital setting, interactions among patients, nurses, and medical doctors can vary significantly due to the differing nature of their relationships (e.g., nurses tend to have only short exchanges with medical doctors while spending more time with patients Vanhems et al. 2013).

Therefore, neglecting community structure in surrogate generation can weaken their realism, as modular organization is known to also affect dynamical processes. Studies have shown that communities influence diffusion by enhancing local spreading while slowing global propagation, leading to different patterns compared to non-modular networks (Salathé and Jones 2010; Nematzadeh et al. 2014; Cencetti et al. 2024), with the spreading speed also being dependent on the network modularity (Peng et al. 2020). Moreover, community-aware measures that incorporate intra and inter-group connections identify influential nodes in such processes more effectively (Rajeh et al. 2021; Blöcker et al. 2022). Following this, without such mesoscopic patterns, surrogate networks risk missing key structural—dynamical relationships present in the original systems.

Consequently, we argue that limitations in characterizing the communities to which interacting individuals belong, represent a critical issue that must be addressed. To this end, we propose an enhancement of ETN-gen by introducing node labels that characterize nodes either based on their metadata, reflecting inherent characteristics when available, or by partitioning them into communities. As a result, a node's ETN will encode not only the structure of its neighborhood, but also information about its community affiliation, as represented by these labels. This refinement is reflected in the generated networks, which exhibit a community organization similar to that of the original network.

Given the scope of this method, we evaluate it by comparing the structural metrics, meaningful for communities, of the starting network, with the generated ones. We do so both at an aggregated level and at the snapshot level, comparing how, in time, the metrics evolve in both networks. We choose several datasets from the *SocioPatterns* study (see section "Datasets") to evaluate our method. Our results show that our methodology is capable of reproducing networks similar to the original ones, significantly improving on the capabilities of the ETN-gen method (Longa et al. 2024) when it comes to the interactions of individuals belonging to different communities and playing different roles. We believe this methodology has broad applicability across various research domains, including epidemiology, urban science, and the study of human interactions in confined settings, such as schools or workplaces.

Related works

Traditional generative models focused primarily on edge dynamics, capturing link appearances and disappearances over time (Karrer and Newman 2009; Perra et al. 2012; Holme 2013; Laurent et al. 2015), but failing to account for higher-order structures that govern real-world interactions. Instead, temporal graphs, especially those modeling human behavior such as face-to-face interactions, demand generative models that can account for both evolving structure and temporal patterns.

Therefore, the use of motifs has been very popular, as they emerge as the key structure to represent local temporal and structural patterns. Motif-based models such as the Structural Temporal Model (STM) (Purohit et al. 2018) estimate motif arrival rates, also using preferential attachment to build aggregated temporal networks, though without controlling for the number of overall motifs and node attributes. DYMOND (Zeno et al.

2021) advances this by modeling both motif frequencies and the roles of nodes within motifs, generating dynamic graphs that reflect realistic structural transitions. However, they only consider few motif structures for static snapshots and only local roles for motifs. Another method using probabilistic modelling on motifs transitions is MTM (Liu and Sariyüce 2023). It reconstructs localized temporal structure effectively but does not model roles or group-level structures, making it insufficient for reproducing emerging communities.

Temporal walk-based methods also aim to capture local dynamics. TagGen (Zhou et al. 2020) performs node-biased temporal walks to capture higher-order structures, while TG-GAN (Zhang et al. 2021) relies on these walks as well as a GAN-based discriminator. Though both capture fine-grained dynamics, they rely heavily on heuristics, lack inductive capabilities, and often duplicate input structures. These approaches have generally been applied to networks with relatively few snapshots and high density, where temporal walks are more likely to revisit relevant nodes and preserve structural coherence. In sparser temporal networks, such as those in our setting, walks are less effective because they tend to drift into low-activity regions, reducing their ability to capture meaningful temporal patterns. TIGGER (Gupta et al. 2022) addresses key limitations in prior models by using intensity-free temporal point processes and a multi-mode inductive decoder to generate interaction graphs with unseen timestamps and arbitrary size. It generalizes well, supports privacy-aware sampling, and demonstrates strong scalability. However, TIGGER does not explicitly model the social or structural roles individuals play over time, nor the emergence of communities.

As for methods that include node attributes in their modelling, CTWalk (Limnios et al. 2022) combines the learning of both temporal walks and node attributes, but it is limited to reproducing discrete content observed in the training data and does not model community structure or meta-roles of nodes. Another method is DYANE (Zeno and Neville 2023), which generates dynamic attributed networks by modeling latent node roles through motif-based participation, jointly producing graph structure and semantic attributes. While it effectively captures local role dynamics, it also specializes in reproducing content of learned motifs, thus it does not explicitly model community structures or the emergence of mesoscale organization, making it less suited for settings where group formation is a central feature.

While prior models focus on motifs, walks, or structural roles, FlowChronicle (Cüppers et al. 2024) introduces a message-passing framework to simulate temporally realistic interaction cascades. Its focus on causal flow and timing precision complements motif-based approaches but does not address community formation or role dynamics, which are central to our work on modeling face-to-face interaction networks.

In contrast to these approaches, our method centers on the temporal evolution of node roles and temporal neighbourhoods as the generative mechanism behind both fine-grained interaction patterns and emergent community structures. This is particularly relevant for face-to-face interaction networks, where individuals move through socially meaningful positions over time. By modeling role dynamics explicitly, our approach captures both the temporal continuity and social organization that underlie group formation-properties that static, walk-based, or motif-counting methods overlook.

Materials and methods

Our methodology, *Labeled ETN-gen* (LETN), builds upon the ETN-gen method, presented in Longa et al. (2024), to account for the interactions that happen among and across communities. We see this network of interactions as a temporal network, in which every snapshot or layer captures all the interactions (links) between study participants (nodes) in a specified interval. We call this interval *gap* and it can be set depending on the type of interactions we have. For example, if the dataset contains face-to-face interactions we can identify meaningful layers grouping interactions with a 5 min interval, while if we take phone call interactions, a whole day would be a more appropriate interval.

As mentioned in the Introduction, we expand ETN-gen by adding a label attribute to the nodes present in the network. In this way, we are able to account for interactions within and between different types of nodes, e.g., students of different classes in a school. In the following subsections we describe the whole methodology and highlight where we applied specific changes to capture roles and communities. A visual illustration of the approach is shown in Figs. 1 and 2.

Identifying patterns

Before the generation of the surrogate networks, we identify repeating patterns of the original network to be reproduced, so that the surrogate ones show the same properties.

Labeled Egocentric Temporal Neighborhood (LETN)

To encode meaningful patterns of interaction, we view networks from an egocentric perspective, that closely follows the evolution of interactions of a specific individual. We follow (Longa et al. 2022, 2024) by firstly identifying an *Egocentric Temporal Neighborhood (ETN)* as all the links the ego node forms in a window of subsequent layers. The window size is identified with a parameter k, for which an optimal value of 2 is obtained

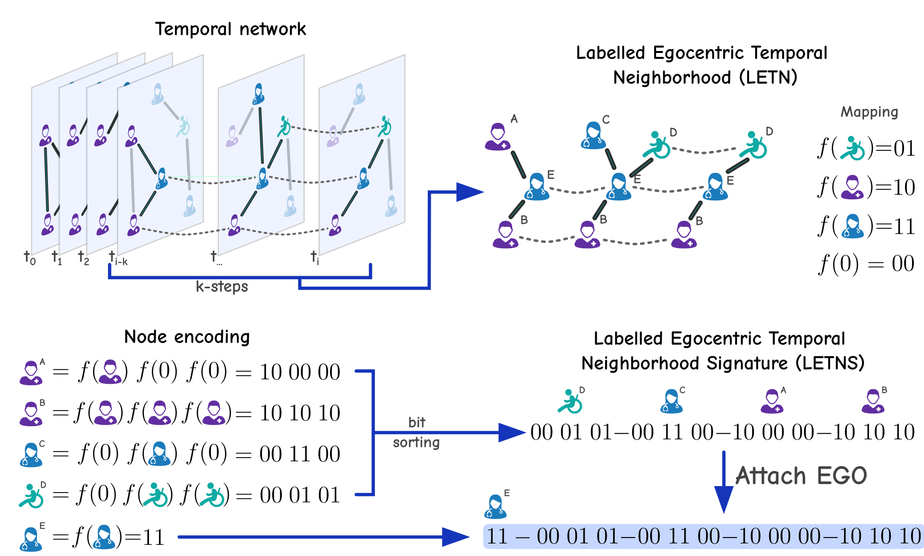

Fig. 1 Labeled Egocentric Temporal Neighborhood Signature (LETNS) encoding. From a window of **k** steps of the original temporal network, we identify the temporal neighborhood of each node from an egocentric perspective. When encoding the signature, differently from the original approach (Egocentric Temporal Neighborhood), we use a binary mapping with 1 when a link from the ego to another node is present in the current snapshot, and 0 when it is not. Finally, we sort the temporal signatures of the neighbors and prepend the resulting string with the node encoding of the ego node to get the complete signature

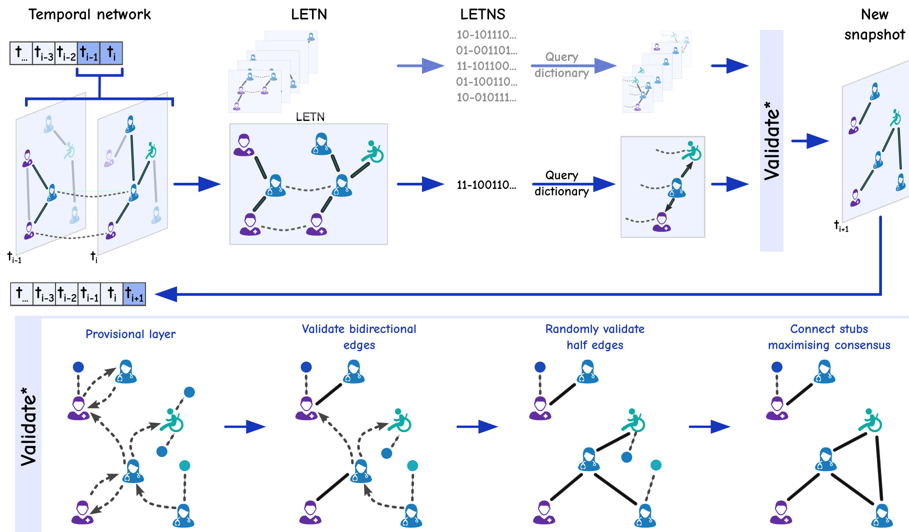

Fig. 2 Generation step for the surrogate network. Starting from $k-1$ graphs as seeds, for each node we consider its LETN and compute the signature, which is used to sample from the probability dictionaries the desired connections in the new layer for that node. After sampling the dictionaries for each of the egos, we validate our *provisional layer* with the displayed rules, which aim at replicating the interactions across communities present in the original network. The new layer can then be added to the temporal network and be used for the generation of the following step

in Longa et al. (2024). As in the previous work, we encode the neighborhood in a unique signature. However, here, nodes are represented using a mapping function encoding node labels into unique binary strings, the *node encoding*. For each node that is connected to the ego node at least once in the window, we represent its temporal signature by concatenating its node encoding in the snapshots where it is connected to the ego node, and using 0 for the snapshots where the link is absent. The following step involves sorting all the temporal signatures of the neighbors in a lexicographic order, so to eliminate dependency on the order in which neighbors are processed. Finally, we add the node encoding of the ego node at the beginning of the string, completing the *Labeled Egocentric Temporal Neighborhood Signature (LETNS)*. Figure 1 illustrates this first step, by showing an example of interaction between a doctor (the ego E), two nurses (A and B), another doctor (C) and a patient (D).

Probability dictionaries

We then use the LETNS to build a dictionary of egocentric temporal neighborhood extensions. Following the original method, for each LETNS we mask the bits corresponding to last temporal layer, i.e., for each neighboring node we only consider its $k-1$ encodings. From the example in Fig. 1, the signature 11 000101 101010 becomes 11 0001x 1010y when masked. These masks are used as keys in a dictionary of temporal extension probability distributions, in which we will store each of the possible extensions, using their counts to compute their distribution. In performing this operation, the original method also uses a *local split* and a *global split*, which allow the generation to capture the periodicity in time. In fact, the method divides all the snapshots in smaller local splits, which repeat over global splits. In most use cases this means considering daily moments, e.g., hours (local split), that repeat across days (global split). Therefore, by grouping all the observations of the local splits across global splits to compute the local distribution, the method stores a dictionary for each local split. In this way we can capture how the individuals have different interactions during the day, and we can also reproduce the periodicity across the global split. Thus, this step is able to model the preferences of nodes with a certain label to interact with nodes with specific labels, also taking into account the time of the interactions.

Surrogate network generation

Once the recurring patterns over local and global splits are identified, it is possible to generate the surrogate networks, following the original methodology.

Provisional layer

The generation steps follow again the original method, but by using the label-dependent dictionaries computed in the previous step. As displayed in Fig. 2, the process starts by using $k-1$ layers as seeds, and extracting the LETNS for each node of the network. In this way, to generate the edges of the following snapshot, or layer, we can sample, with the current neighborhood signature, from the probability dictionaries defined above to obtain layer k based on the possible extensions. This will determine a *provisional layer*, in which each node has some desired connections to nodes with specific labels. Once the provisional layer has been validated (see next paragraph), the new snapshot is generated, added in the new network, and used for the generation of the following step. The steps can be repeated until the desired length of the network is reached.

Connection validation

The validation step applies a series of rules needed to reach a consensus among the links required by all the ego nodes when choosing their neighborhood. This step differs from Longa et al. (2024) as the rules take into account the node labels. The validation, graphically shown in Fig. 2, consists in:

1. Validate the bidirectional edges, which represent the reciprocal requests (node i requires a connections with node j and node j with node i). Since the two nodes agree to be connected the link is accepted. This maximizes the adherence of the final layer to the sampled LETN.
2. Randomly validate half of the unidirectional edges, i.e., the unreciprocal requests (node i requires a connection with node j but node j does not require the connection with node i).
3. Connect the remaining stubs maximizing the consensus, i.e., maximizing the matching of the remaining reciprocal requests.

Label absence

If nodes labels are not provided as metadata with the temporal network dataset, we can assign labels based on the network community structure.

A first strategy consists in considering the aggregated network obtained by collapsing all the layers in one static network (where links are weighted according to the number of times that they appeared in the temporal layers), and to apply a community detection algorithm. The labels can then be assigned according to the obtained communities. We chose to use the Louvain algorithm (Blondel et al. 2008), but other choices, like the Leiden algorithm (Traag et al. 2019), are possible. We call this strategy *Community-LETN (CLETN)*.

A second strategy, which we call *Dynamic-LETN (DLETN)*, is instead useful when the communities change significantly during the observation period. This happens, for instance, for the school datasets, since communities during classes are different than communities during breaks. The strategy consists in aggregating at the local split level, thus obtaining multiple aggregated networks instead of only one. The community detection can then be applied to every aggregated network, obtaining different partitions for each local split. In this approach, the probability dictionaries will use these different partitions for each local split, resulting in different dictionaries for each local split.

Datasets

To ascertain the validity of the synthetic networks we generate, we select 7 datasets from the SocioPatterns repository (see Data Availability for details on how to access them). They contain data about face-to-face interactions over different period of times in different environments. Specifically, we select one dataset corresponding to a primary school (Stehlé et al. 2011; Gemmetto et al. 2014), three for a high school (Fournet and Barrat 2014; Mastrandrea et al. 2015), one for a hospital (Vanhems et al. 2013) and two for an office workplace (Génois et al. 2015; Génois and Barrat 2018). Each of these datasets contains metadata, setting the nodes label: there are classes and teachers for the schools, doctors and patients and nurses for the hospital, and office roles for the workplace datasets. This allowed us to test the base method and also the CLETN and DLETN extensions. The following Table 1 shows descriptive statistics of the used datasets, in which we see how the chosen sample has different ratios of labels and participants and data collection duration and number of interactions.

In order to avoid possible biases in specific generations, for each of these datasets we generated 10 synthetic networks, starting from snapshots of the original graph as seeds, and then changing the seed for the dictionary sampling and validation step, and averaged the results. For these generations we used a window of $k = 2$ and $gap = 300$, i.e., snapshots of 5 min are used to group interactions.

Computational complexity

We analyze here the computational complexity of the three main components of the LETN framework: signature computation, counting signatures for the probability dictionaries over the input network, and surrogate network generation. Each step has distinct characteristics and performance implications depending on the network size, temporal resolution, and structural sparsity.

LETNS computation

The worst-case complexity of computing the LETN signature for a given node and time step depends on two main steps: constructing a binary encoding for each neighbor

Table 1 Statistical description of the datasets. The table lists number of participants, unique labels in the original metadata, length of the data collection (in days and hours) and total number of interactions

Dataset	Participants	Unique labels	Length	Interactions
Primary school	242	6	1 d 9 h	125773
Highschool 2011	126	4	3 d 4 h	28561
Highschool 2012	180	5	8 d 11 h	45047
Highschool 2013	327	9	4 d 5 h	188508
Hospital	75	4	4 d	32424
Workplace 2013	92	5	11 d 10 h	9827
Workplace 2015	217	12	11 d 12 h	78249

in the temporal neighborhood, and sorting all encodings to form the final signature. Each encoding corresponds to a binary string of length $(k+1) \cdot |f(\ell(v'))|$, where $\ell(v')$ is the label of neighbor v', and f maps labels to binary strings of fixed size. Since the label space is much smaller than the number of nodes, $|f(\ell(v'))|$ is constant. Letting $d^{(k)}$ denote the maximum number of neighbors a node can have within a temporal window of size $k \cdot \Delta t$ (i.e., the temporal neighbourhood), where Δt is the temporal gap, the cost of encoding is $\mathcal{O}(d^{(k)})$, and the cost of sorting is $\mathcal{O}(d^{(k)} \log d^{(k)})$. Therefore, the total worst-case complexity per signature is:

$$\mathcal{O}(d^{(k)} \log d^{(k)})$$

LETNS counting

The worst-case complexity of counting all LETNS across a temporal network depends on iterating over each node and each snapshot, extracting the corresponding local neighborhood, computing the signature, and updating the count statistics. Let n be the number of nodes, and m the number of temporal snapshots, computed as $m = (T_{\text{end}} - T_{\text{start}})/\Delta t$, where T_{start} and T_{end} denote the time range covered by the network, and Δt is the temporal gap. For each of the $n \cdot m$ node-time pairs, the cost of extracting the temporal neighborhood is upper bounded by $\mathcal{O}(d^{(k)} \cdot k)$, and the cost of computing and sorting the signature is $\mathcal{O}(d^{(k)} \log d^{(k)})$, as previously discussed. Since signature matching and count updates can be performed in constant time $\mathcal{O}(1)$, the total worst-case complexity becomes:

$$\mathcal{O}(n \cdot m \cdot d^{(k)} \log d^{(k)})$$

In practice, for reasonable values of k and Δt, the term $d^{(k)}$ remains bounded independently of the overall network size, resulting in an effective complexity of $\mathcal{O}(n \cdot m)$.

Surrogate network generation

The computational cost of generating a surrogate network using LETN depends on iteratively sampling and validating local patterns across all nodes and time steps. At each generation step, a node-time pair is selected, and a corresponding signature is sampled from the learned distribution. This operation has negligible cost due to the compact size of the dictionary. The sampled signature is then decoded to reconstruct the corresponding local temporal neighborhood, which involves adding at most $d^{(k)}$ edges based on the encoded structure. Since this decoding process is repeated for each of the $n \cdot m$ node-time pairs, the overall generation complexity is:

$$\mathcal{O}(n \cdot m \cdot d^{(k)})$$

Assuming $d^{(k)}$ remains bounded for reasonable values of k and Δt, the generation procedure scales linearly with the number of nodes and time steps. Unlike the counting phase, no sorting or encoding is required during generation, which makes the process significantly faster in practice.

Results

Our analyses aim to prove that the addition of labels in the ETN-gen methodology allows to capture the structure of interactions among and across communities, while the original ETN-gen approach misses these structures. Thus, the first step in this direction is visualizing how interaction neighborhoods encoded by LETN are more indicative of communities than the ones encoded by ETN. To do so, we identify unique neighborhood signatures, using both ETN signatures and LETN signatures, across all nodes and temporal layers for the High School 2011 original network. Then, for each node, we count the occurrence of each of the signatures found by the two methods, thus generating a vector per node. After this, we apply Principal Component Analysis (PCA) (Jolliffe and Cadima 2016) to visualize whether nodes of similar communities have similar structures of neighborhood. This is indeed the case, as shown in Fig. 3. In the figure we firstly visualize, with the same technique, the degree of nodes, showing how the task is not trivial. Then, we show the outcome of PCA for the original network when the patterns are identified by either ETNS or LETNS. It is clear that, using LETNS, we can capture community patterns with the neighborhoods that it identifies, displayed by the fact that nodes belonging to the same class form clear clusters in the visualization. On the contrary, ETNS is not able to capture these dynamics. An interesting observation here, is that the behavior of teachers, who, differently from students, typically move across classes rather then interacting between themselves, is captured as well. Following this, we visualize the improvements on the base methodology on the High School 2013 dataset, as it presents more classes, and thus labels, to test the validity of the model.

Moreover, we compare our proposed approach against one competitor, namely *MTM* (Liu and Sariyüce 2023), which is motif-based. While other methods exist that incorporate node attributes, they are primarily deep learning models relying on random walks. As discussed in section "Related works", such approaches are better suited to denser networks with fewer snapshots, where short walks from a limited set of nodes can still capture local structure. In our sparser setting with many snapshots, random walks fail to efficiently reconstruct the global structure needed for generating realistic surrogates.

Figure 4 shows in three panels how the method we are proposing is able to capture communities better than the competitors. First of all, we show the original network, a LETN generated network, a ETN generated network and a MTM one, by aggregating all the temporal layers of the network, thus linking nodes that have at least one interaction during the measurement period. It is evident how networks obtained with the LETN method are characterized by a larger quantity of links inside the communities, resembling more the original network than ETN and MTM, where instead nodes are linked independently on the communities. Panel **B** of Fig. 4 shows more in detail how the interactions are distributed among and across communities, by using heatmaps

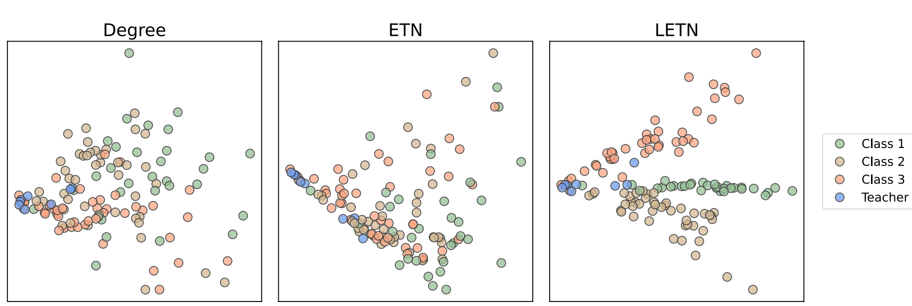

Fig. 3 PCA visualization of the space of neighborhoods of the High School 2011 dataset. On the left we have the visualization of the degree counts via PCA, showing how differentiating nodes is not a trivial task. Comparing the visualizations for ETN and LETN generated networks, we see that the neighborhoods identified by ETN are not sufficient to distinguish clusters of nodes that have a similar behavior. With LETN, instead, we are able to construct clusters of the three classes

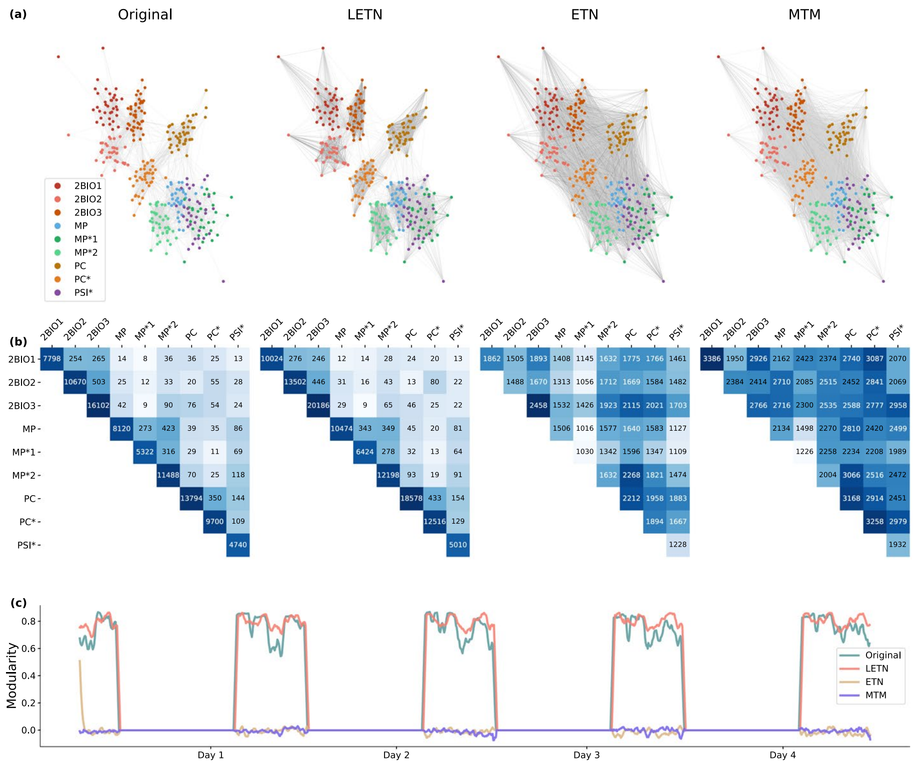

Fig. 4 Comparison of the original network with networks generated by LETN, ETN and MTM, for the High School 2013 dataset. Panel **A** shows the networks aggregated over the whole measurement period. LETN generates a network with many interactions inside communities, and less outside communities, better reproducting the original network than the other methods. The number of interactions are shown in Panel **B**, where columns and rows represent the school classes. The original patterns are reproduced by LETN, but not by ETN nor MTM. The final Panel **C** shows modularity through time (set to 0 when there were no interactions). Following the above observations, ETN and MTM have a poor modularity, while LETN closely follows the original network's trend

(log-scaled colors). We also see, once again, how LETN is able to well reproduce how the interactions are distributed across communities. ETN and MTM, instead, are not able to capture any such pattern, as both are limited by not including labels in the patterns they model. The final Panel **C**, shows the modularity (original labels are used), which is a proxy of the goodness of network partitions, over the measurement period. For visualization purposes, we set modularity to 0 when there were no interactions recorded, i.e., hours not at school. This step also shows the characteristic of the base method to capture interactions periodicity in the generated networks. We see that ETN is not able to generate a high modularity (the starting value is due to the seed graphs used in the generation), which always stays close to 0, meaning that links among and across communities are random and do not follow a fixed structure. The results from MTM are similar. Instead, LETN closely follows the trend of the original network's modularity.

Another improvement of the LETN method consists in the ability of our method to better reproduce the duration of the interactions among individuals of different communities. For this purpose, we have computed, for each interaction, the number of consecutive snapshots in which it is present. Then, we aggregate these computed lengths over the communities the nodes belong to, and average them, obtaining the mean duration of an interaction between nodes depending on their community. The result is shown in Fig. 5, which reports the average number of snapshots that an interaction between two specific communities lasts. We see an interesting behavior in the matrix representing the

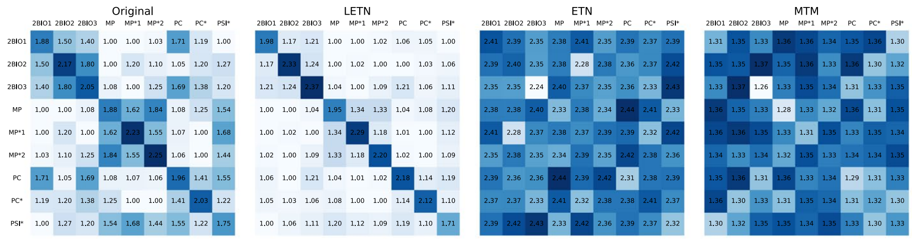

Fig. 5 Mean number of consecutive interactions across communities, i.e., classes, of the High School 2013 dataset. The heatmaps show, on average, how many consecutive snapshots does an interaction between two communities last. We see how LETN, with the use of labels, improves the ETN-based approach and outperforms MTM in reproducing the average duration of the face-to-face interactions

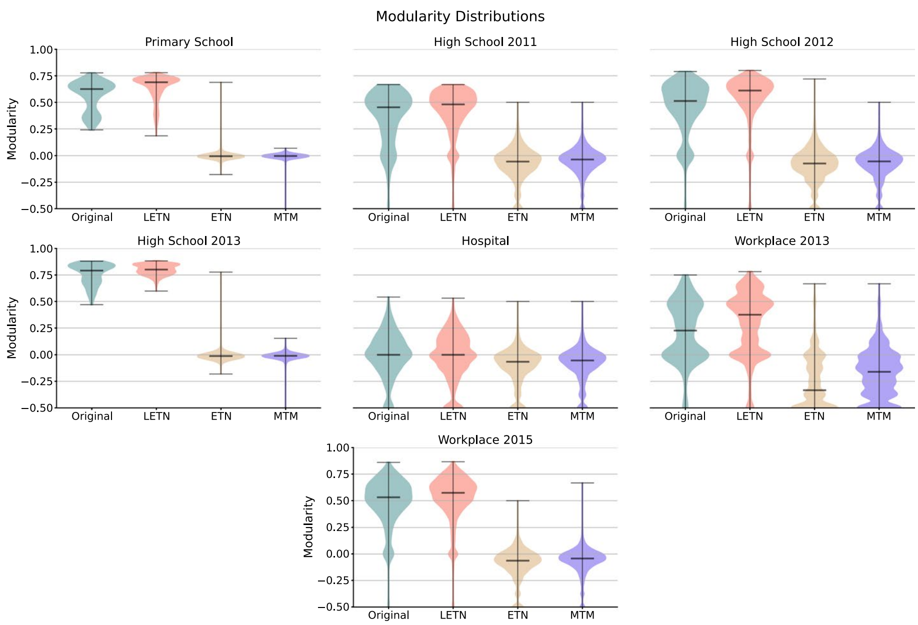

Fig. 6 Distributions of modularity across all datasets, for the original network and the ones generated by LETN, ETN and MTM (the middle line is the median). We see how the generation of LETN is able to produce distributions very similar to the original ones, while the other methods are not

original dataset, namely that not only long interactions are present inside the same community (diagonal of the matrix), but that clusters are formed around classes that belong to the same subject. Therefore, it is meaningful that the LETN method is able to capture these patterns, while ETN, overestimating the number of interactions, also overestimates the average length of interactions and is not able to model any of these patterns. Even MTM, which models motifs and their arrival times, fails in reproducing the original durations of interactions, unlike our method.

The outcomes presented for the High School 2013 dataset are similar to the ones that we obtain for all the other datasets we considered. Figures 6 and 7 summarize this by representing the distribution of *modularity* and *label assortativity* for the original datasets, and for the synthetic networks generated by LETN, ETN and MTM. It is evident how the modularity distributions of the original datasets are well reproduced by the LETN method, while the other methods fail. It is interesting noting that, both competitors being based only on motifs and local patterns, they fail in a similar way. Similar results are obtained for label assortativity. These results are a validation of how we are

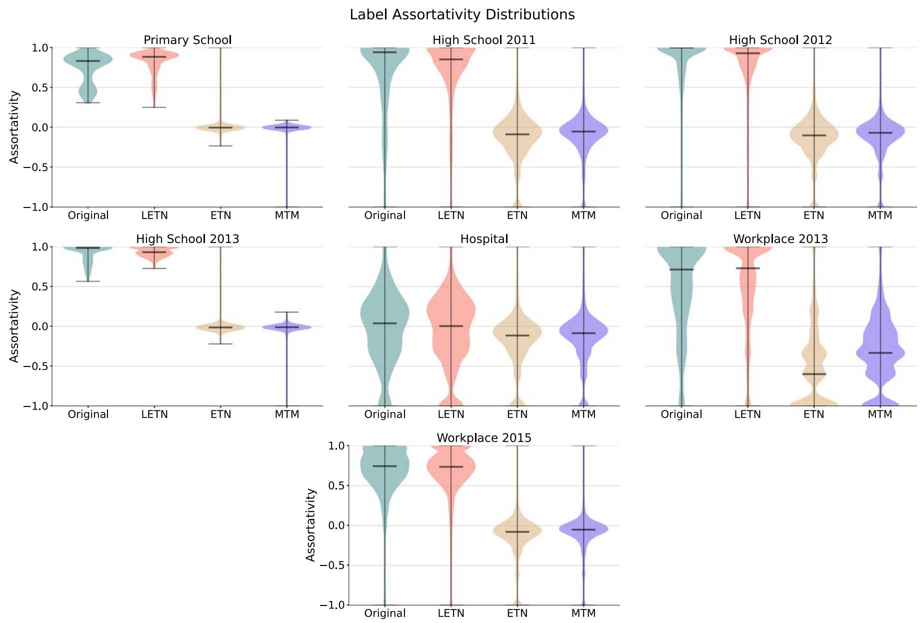

Fig. 7 Distributions of label assortativity across all datasets, for the original network and the ones generated by LETN, ETN and MTM (the middle line is the median). We see how the generation of LETN is able to capture the interactions between nodes with different labels, while ETN and MTM cannot

able to capture structures of interactions which depend on the type (e.g., role) of the nodes that participate in those interactions, showing how the node characteristics are relevant when generating a surrogate network, and that modelling local patterns is not sufficient to capture community dynamics.

All the shown results correspond to $gap = 5$ min, which has resulted to be the most suited for the identification of interactions. However results do not change when we use $gap = 15$ min, as reported in Table 2, which shows the average modularity and label assortativity over the measurement period (ten generations per method). In the same table we also report the results obtained for the aggregated networks (i.e., when all the temporal layers are collapsed to form one static network).

Likewise, all the results set $k = 2$ as the number of snapshots to compute the temporal neighbourhood, as that was the optimal value proposed in the original ETN-gen paper. To evaluate the difference, we compare *modularity*, *label assortativity* and *clustering coefficient* of each snapshot for generated networks with values of *k* from 2 to 5. Then, we compute the euclidean distance of these sequences from the sequence obtained for the original network, and compare the obtained distances, for each dataset. Table 3 shows the outcome, in which we see that the difference across different values of *k* is marginal. Thus, overall we confirm that using $k = 2$ results in surrogated networks closer to the original.

Finally, we validate the two strategies proposed to deal with the absence of labeled data, namely CLETN, which assigns labels based on the partitions found on the aggregated graph, and DLETN, which assigns labels based on the partitions computed at the local split level. To do so, we compare the trend of modularity and label assortativity over all the snapshots between their generated networks and the original one. In particular, we compute *modularity*, *label assortativity* and *clustering coefficient* for each snapshot of the original network and the generated ones, then, we compute a euclidean

Table 2 Average modularity and label assortativity of all the methods with gaps of 5 min, 15 min and the aggregated graph

Dataset	Method	Modularity			Label assortativity		
		5 min	15 min	Aggr.	5 min	15 min	Aggr.
Primary school	Original	0.56 ± 0.14	0.57 ± 0.14	0.59	0.74 ± 0.19	0.71 ± 0.20	0.31
	LETN	0.64 ± 0.11	0.65 ± 0.10	0.69 ± 0.01	0.83 ± 0.13	0.65 ± 0.10	0.52 ± 0.01
	ETN	0.00 ± 0.06	0.00 ± 0.04	0.66 ± 0.01	0.00 ± 0.08	0.00 ± 0.04	0.01 ± 0.00
Highschool 2011	Original	0.37 ± 0.24	0.37 ± 0.25	0.56	0.75 ± 0.40	0.68 ± 0.43	0.65
	LETN	0.42 ± 0.15	0.44 ± 0.14	0.56 ± 0.01	0.77 ± 0.19	0.44 ± 0.14	0.58 ± 0.01
	ETN	−0.08 ± 0.08	−0.06 ± 0.06	0.48 ± 0.01	−0.13 ± 0.16	−0.06 ± 0.06	0.00 ± 0.01
Highschool 2012	Original	0.46 ± 0.24	0.50 ± 0.20	0.66	0.89 ± 0.26	0.89 ± 0.21	0.63
	LETN	0.55 ± 0.15	0.56 ± 0.14	0.67 ± 0.01	0.84 ± 0.16	0.56 ± 0.14	0.60 ± 0.01
	ETN	−0.09 ± 0.07	−0.07 ± 0.06	0.55 ± 0.01	−0.14 ± 0.14	−0.07 ± 0.06	0.01 ± 0.00
Highschool 2013	Original	0.76 ± 0.10	0.77 ± 0.08	0.80	0.92 ± 0.11	0.91 ± 0.10	0.65
	LETN	0.80 ± 0.04	0.80 ± 0.04	0.81 ± 0.01	0.93 ± 0.05	0.80 ± 0.04	0.62 ± 0.01
	ETN	−0.01 ± 0.06	0.00 ± 0.06	0.77 ± 0.01	−0.01 ± 0.07	0.00 ± 0.06	0.01 ± 0.01
Hospital	Original	0.00 ± 0.20	−0.01 ± 0.20	0.14	−0.03 ± 0.43	−0.08 ± 0.40	−0.12
	LETN	−0.02 ± 0.16	−0.02 ± 0.17	0.13 ± 0.01	−0.07 ± 0.35	−0.02 ± 0.17	−0.13 ± 0.01
	ETN	−0.11 ± 0.09	−0.10 ± 0.09	0.10 ± 0.01	−0.20 ± 0.21	−0.10 ± 0.09	−0.01 ± 0.01
Workplace 2013	Original	0.22 ± 0.28	0.33 ± 0.25	0.55	0.55 ± 0.56	0.65 ± 0.40	0.52
	LETN	0.29 ± 0.12	0.40 ± 0.11	0.55 ± 0.01	0.60 ± 0.18	0.40 ± 0.11	0.41 ± 0.01
	ETN	−0.25 ± 0.08	−0.16 ± 0.07	0.42 ± 0.01	−0.54 ± 0.14	−0.16 ± 0.07	−0.01 ± 0.01
Workplace 2015	Original	0.50 ± 0.21	0.52 ± 0.20	0.57	0.71 ± 0.26	0.71 ± 0.26	0.32
	LETN	0.54 ± 0.12	0.55 ± 0.13	0.62 ± 0.01	0.70 ± 0.10	0.55 ± 0.13	0.30 ± 0.01
	ETN	−0.08 ± 0.08	−0.07 ± 0.07	0.40 ± 0.01	−0.13 ± 0.14	−0.07 ± 0.07	0.00 ± 0.01

distance on the resulting vectors. In this way we compare how much the generation of both LETN, CLETN, and DLETN follow the trend of the original network when it comes to these structural measures. From the results in Table 4, it is clear that both the two proposed extensions do not differ significantly from the results of the main LETN methodology, thus proving to be valid alternatives when metadata with node characteristics are not available.

Discussion

The method we propose allows us to generate temporal networks that serve as surrogates to observed real-world datasets. Our approach builds on ETN-gen (Longa et al. 2024), by adding a mechanism of node labeling, that has as a main advantage to reproduce the organization of nodes in communities, i.e., groups of nodes highly connected among them and less connected to the other groups of nodes. Communities can be observed in any kind of datasets (Leskovec et al. 2008) but they are particularly relevant for social networks, in some cases spontaneously emerging as in groups of friends or colleagues (Isella et al. 2011; Génois and Barrat 2018), and in some cases imposed by the environment setting, as in schools where students are usually separated into classes (Stehlé et al. 2011). The structure created by the communities is not poorly relevant since it has a major effect on the global phenomena that can be observed on a network. Examples are represented by spreading phenomena, where the effect of communities is to enhance the local spreading while slowing down the global one (Salathé and Jones 2010; Cencetti et al. 2024), thus resulting in a completely different diffusion pattern with respect to networks without a community structure. Similar effects can be observed in

Table 3 Euclidean distance for the selected metrics for different values of the *k* parameter. We compute the metrics at each snapshot for the original data and the networks generated with different *k* values, then compute their euclidean distance. Here, we report average and error for each metric

Dataset	k value	Modularity	Label assort.	Clustering coeff.
Primaryschool	2	3.33 ± 0.04	4.21 ± 0.05	1.62 ± 0.01
	3	3.35 ± 0.06	4.22 ± 0.08	1.98 ± 0.02
	4	3.48 ± 0.10	4.49 ± 0.12	2.06 ± 0.01
	5	3.51 ± 0.04	4.58 ± 0.05	2.11 ± 0.01
Highschool 2011	2	6.45 ± 0.17	11.78 ± 0.21	0.79 ± 0.03
	3	6.37 ± 0.10	11.49 ± 0.25	0.83 ± 0.01
	4	6.46 ± 0.14	11.75 ± 0.16	0.84 ± 0.01
	5	6.57 ± 0.12	11.92 ± 0.30	0.86 ± 0.01
Highschool 2012	2	13.63 ± 0.10	20.17 ± 0.15	0.76 ± 0.01
	3	13.71 ± 0.15	20.34 ± 0.22	0.77 ± 0.01
	4	13.67 ± 0.13	20.46 ± 0.19	0.78 ± 0.00
	5	13.71 ± 0.13	20.88 ± 0.23	0.78 ± 0.00
Highschool 2013	2	3.81 ± 0.04	4.35 ± 0.05	1.11 ± 0.01
	3	4.51 ± 0.03	5.21 ± 0.04	1.20 ± 0.01
	4	5.11 ± 0.03	5.97 ± 0.04	1.22 ± 0.00
	5	5.64 ± 0.04	6.61 ± 0.04	1.23 ± 0.01
Hospital	2	7.09 ± 0.14	13.79 ± 0.29	1.48 ± 0.01
	3	7.78 ± 0.19	15.04 ± 0.41	1.58 ± 0.02
	4	8.42 ± 0.18	16.11 ± 0.31	1.63 ± 0.01
	5	8.97 ± 0.20	17.09 ± 0.39	1.65 ± 0.01
Workplace 2013	2	15.01 ± 0.28	27.06 ± 0.41	0.36 ± 0.00
	3	15.02 ± 0.17	27.21 ± 0.36	0.36 ± 0.00
	4	15.15 ± 0.20	27.46 ± 0.43	0.36 ± 0.00
	5	15.14 ± 0.14	27.53 ± 0.25	0.36 ± 0.00
Workplace 2015	2	13.98 ± 0.27	19.34 ± 0.29	0.82 ± 0.00
	3	14.04 ± 0.28	19.62 ± 0.42	0.82 ± 0.00
	4	14.17 ± 0.25	19.92 ± 0.39	0.83 ± 0.00
	5	14.19 ± 0.18	19.91 ± 0.32	0.83 ± 0.00

opinion dynamics (Peng et al. 2023), information transfer (Danon et al. 2008), and synchronization (Arenas et al. 2008).

In general, the existence of structure in a network is what distinguishes it from a random network, hence adding communities to surrogate networks makes them more similar to the original ones and, thus, more realistic. The ability of our method to reproduce the modular organization, interaction patterns, and temporal characteristics of the original networks is also consistent with previous findings on the role of mesoscopic structures in shaping dynamics. Studies have shown that community-aware centrality measures capturing both intra and inter-community flows (Rajeh et al. 2021; Blöcker et al. 2022) identify influential nodes more effectively, and that diffusion efficiency has to balance local reinforcement and global reach, thus following community structures (Nematzadeh et al. 2014). Modularity has also been linked to optimal balancing of local and global interactions for diffusion processes (Galstyan and Cohen 2007; Peng et al. 2020). Therefore, by preserving the key structural features, our surrogates are not only structurally faithful but may also reproduce the characteristic spreading patterns, activation timings, and influence distributions described in the literature. This suggests that they could serve as reliable testbeds for investigating how structural roles and community organization impact dynamical processes.

Table 4 Euclidean distance for the selected metrics for LETN and the CLETN and DLETN extensions. We compute the metrics at each snapshot for the original data and the networks generated by our method. Then, we compute the euclidean distance between the original sequence and each of the generated ones. Here, we report average and error over 10 generations for each method. The very close results confirm the validity of the extensions we propose for LETN

Dataset	Method	Modularity	Label assort.	Clustering coeff.
Primary School	LETN	3.21 ± 0.05	4.07 ± 0.06	1.72 ± 0.02
	CLETN	3.07 ± 0.05	3.93 ± 0.06	1.55 ± 0.01
	DLETN	3.21 ± 0.06	4.06 ± 0.08	1.43 ± 0.02
Highschool 2011	LETN	6.45 ± 0.17	11.78 ± 0.21	0.79 ± 0.03
	CLETN	6.07 ± 0.17	11.14 ± 0.40	0.80 ± 0.02
	DLETN	5.69 ± 0.20	10.69 ± 0.40	0.80 ± 0.02
Highschool 2012	LETN	13.63 ± 0.10	20.17 ± 0.15	0.76 ± 0.01
	CLETN	11.40 ± 0.14	18.28 ± 0.19	0.76 ± 0.01
	DLETN	12.63 ± 0.23	19.61 ± 0.33	0.76 ± 0.01
Highschool 2013	LETN	3.81 ± 0.04	4.35 ± 0.05	1.11 ± 0.01
	CLETN	3.87 ± 0.04	4.87 ± 0.05	1.11 ± 0.01
	DLETN	3.64 ± 0.03	4.39 ± 0.04	1.11 ± 0.01
Hospital	LETN	7.09 ± 0.14	13.79 ± 0.29	1.48 ± 0.01
	CLETN	7.86 ± 0.18	15.09 ± 0.39	1.56 ± 0.01
	DLETN	7.47 ± 0.18	14.41 ± 0.35	1.56 ± 0.01
Workplace 2013	LETN	15.01 ± 0.28	27.06 ± 0.41	0.36 ± 0.01
	CLETN	14.26 ± 0.16	26.19 ± 0.25	0.36 ± 0.01
	DLETN	14.25 ± 0.19	26.17 ± 0.38	0.36 ± 0.01
Workplace 2015	LETN	13.98 ± 0.27	19.34 ± 0.29	0.82 ± 0.01
	CLETN	12.83 ± 0.32	18.74 ± 0.45	0.82 ± 0.01
	DLETN	13.14 ± 0.28	19.43 ± 0.48	0.82 ± 0.01

In our method, the community structure is not imposed as a global or meso-scale feature, but it emerges from the microscopic dynamics, being incorporated in the local development rules and in how each node chooses its neighbors. The ego-centric perspective also allows us to distinguish among nodes belonging to different categories. If categories are characterized by different individual behaviors in the original network, these differences are naturally reproduced in the surrogate networks, once again tying to the importance of specific nodes, which might be influenced by their category as well. In the hospital dataset, for instance, we can reproduce the fact that nurses are characterized by a higher number of interactions, especially with patients and among themselves, while medical doctors have less interactions but with a longer duration. This mechanism increases the nodes heterogeneity, a peculiarity of social networks that the previous method, ETN-gen, could not reproduce.

With our method we showed how an already efficient method like ETN-gen can be further improved by adding node labels. This represents only a little increase in the algorithm complexity with significant positive implications. By using it, we are in fact able to reproduce interactions with a higher fidelity to the original network, both considering the distribution of links among nodes and their temporal duration. Moreover, we fill a gap in the literature, for which most of the methods generating attributed temporal networks, focus on dense networks with few snapshots, with their random walk approaches not being suited to the generation of surrogate networks.

This method can be applied in the cases where communities are provided by metadata or not, as in the second case they can be obtained by a simple community detection algorithm (Girvan and Newman 2002), which can be applied to the aggregated network or even to each singular snapshot of the temporal network.

In conclusion, we believe that this lightweight method can be a useful tool for the study of human interactions, particularly in the case of face-to-face interactions.

Abbreviations

ETN	Egocentric Temporal Neighbourhood
LETN	Labeled Egocentric Temporal Neighbourhood
ETNS	Egocentric Temporal Neighbourhood Signature
LETNS	Labeled Egocentric Temporal Neighbourhood Signature
CLETN	Community Labeled Egocentric Temporal Neighbourhood
DLETN	Dynamic Labeled Egocentric Temporal Neighbourhood

Author contributions

All authors designed the project. N.A.G., A.L. and G.T. implemented the model. N.A.G. and A.L. run the experiments. N.A.G., A.L. and G.C. wrote the main manuscript. All authors reviewed the manuscript.

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Code and data availibility

All the dataset used in the analysis are publicly available on the SocioPatterns project website: http://www.sociopatterns.org. The code can be found at the following GitHub repository: https://github.com/Svidon/LETN-gen.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

Aharony N, Pan W, Ip C, Khayal I, Pentland A (2011) Social fMRI: investigating and shaping social mechanisms in the real world. Pervasive Mob Comput 7(6):643–659. https://doi.org/10.1016/j.pmcj.2011.09.004

Alessandretti L, Aslak U, Lehmann S (2020) The scales of human mobility. Nature 587(7834):402–407. https://doi.org/10.1038/s41586-020-2909-1

Arenas A, Díaz-Guilera A, Kurths J, Moreno Y, Zhou C (2008) Synchronization in complex networks. Phys Rep 469(3):93–153. https://doi.org/10.1016/j.physrep.2008.09.002

Arregui-García B, Longa A, Lotito QF, Meloni S, Cencetti G (2024) Patterns in temporal networks with higher-order egocentric structures. Entropy 26(3):256. https://doi.org/10.3390/e26030256

Arrell D, Terzic A (2010) Network systems biology for drug discovery. Clin Pharmacol Ther 88(1):120–125. https://doi.org/10.1038/clpt.2010.91

Barabasi A-L (2016) Network science. Cambridge University Press, Cambridge

Bassett DS, Sporns O (2017) Network neuroscience. Nat Neurosci 20(3):353–364. https://doi.org/10.1038/nn.4502

Batty M (2013) The new science of cities. The MIT Press, Cambridge. https://doi.org/10.1080/13658816.2014.937717

Beard C, Millner AJ, Forgeard MJ, Fried EI, Hsu KJ, Treadway MT, Leonard CV, Kertz S, Björgvinsson T (2016) Network analysis of depression and anxiety symptom relationships in a psychiatric sample. Psychol Med 46(16):3359–3369. https://doi.org/10.1017/S0033291716002300

Blöcker C, Nieves JC, Rosvall M (2022) Map equation centrality: community-aware centrality based on the map equation. Appl Netw Sci 7(1):56. https://doi.org/10.1007/s41109-022-00477-9

Blondel VD, Guillaume J-L, Lambiotte R, Lefebvre E (2008) Fast unfolding of communities in large networks. J Stat Mech: Theory Exp 2008(10):10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Bontorin S, Cencetti G, Gallotti R, Lepri B, De Domenico M (2024) Emergence of complex network topologies from flow-weighted optimization of network efficiency. Phys Rev X 14(2):021050. https://doi.org/10.1103/PhysRevX.14.021050

Borsboom D, Deserno MK, Rhemtulla M, Epskamp S, Fried EI, McNally RJ, Robinaugh DJ, Perugini M, Dalege J, Costantini G et al (2021) Network analysis of multivariate data in psychological science. Nat Rev Methods Primers 1(1):58. https://doi.org/10.1038/s43586-021-00055-w

Cattuto C, Broeck W, Barrat A, Colizza V, Pinton J-F, Vespignani A (2010) Dynamics of person-to-person interactions from distributed rfid sensor networks. PLoS ONE 5(7):e11596. https://doi.org/10.1371/journal.pone.0011596

Cencetti G, Barrat A (2024) Generating surrogate temporal networks from mesoscale building blocks. arXiv:2411.05477

Cencetti G, Lucchini L, Santin G, Battiston F, Moro E, Pentland A, Lepri B (2024) Temporal clustering of social interactions trades-off disease spreading and knowledge diffusion. J R Soc Interface 21(210):20230471. https://doi.org/10.1098/rsif.2023.0471

Centellegher S, De Nadai M, Caraviello M, Leonardi C, Vescovi M, Ramadian Y, Oliver N, Pianesi F, Pentland A, Antonelli F et al (2016) The mobile territorial Lab: a multilayered and dynamic view on parents' daily lives. EPJ Data Sci 5:1–19. https://doi.org/10.1140/epjds/s13688-016-0064-6

Cüppers J, Schoen A, Blanc G, Gimenez P-F (2024) Flowchronicle: synthetic network flow generation through pattern set mining. Proc ACM Netw 2(CoNEXT4):1–20. https://doi.org/10.1145/3696407

Dall'Amico L, Barrat A, Cattuto C (2024) An embedding-based distance for temporal graphs. Nat Commun 15(1):9954

Danon L, Arenas A, Diaz-Guilera A (2008) Impact of community structure on information transfer. Phys Rev E Stat Nonlinear Soft Matter Phys 77(3):036103. https://doi.org/10.1103/PhysRevE.77.036103

Eagle N, Pentland A (2006) Reality mining: sensing complex social systems. Pers Ubiquit Comput 10:255–268. https://doi.org/10.1007/s00779-005-0046-3

Eagle N, Pentland AS (2009) Eigenbehaviors: identifying structure in routine. Behav Ecol Sociobiol 63:1057–1066. https://doi.org/10.1007/s00265-009-0739-0

Fournet J, Barrat A (2014) Contact patterns among high school students. PLoS ONE 9(9):107878. https://doi.org/10.1371/journal.pone.0107878

Galdeman A, Zignani M, Gaito S (2023) Unfolding temporal networks through statistically significant graph evolution rules. In: 2023 IEEE 10th international conference on data science and advanced analytics (DSAA). IEEE, pp 1–10

Galstyan A, Cohen P (2007) Cascading dynamics in modular networks. Phys Rev E Stat Nonlinear Soft Matter Phys 75(3):036109. https://doi.org/10.1103/PhysRevE.75.036109

Gelardi V, Le Bail D, Barrat A, Claidiere N (2021) From temporal network data to the dynamics of social relationships. Proc R Soc B 288(1959):20211164. https://doi.org/10.1098/rspb.2021.1164

Gemmetto V, Barrat A, Cattuto C (2014) Mitigation of infectious disease at school: targeted class closure vs school closure. BMC Infect Dis 14:1–10. https://doi.org/10.1186/s12879-014-0695-9

Génois M, Barrat A (2018) Can co-location be used as a proxy for face-to-face contacts? EPJ Data Sci 7(1):1–18. https://doi.org/10.1140/epjds/s13688-018-0140-1

Génois M, Vestergaard CL, Fournet J, Panisson A, Bonmarin I, Barrat A (2015) Data on face-to-face contacts in an office building suggest a low-cost vaccination strategy based on community linkers. Netw Sci 3(3):326–347. https://doi.org/10.1017/nws.2015.10

Girardini NA, Centellegher S, Passerini A, Bison I, Giunchiglia F, Lepri B (2023) Adaptation of student behavioural routines during Covid-19: a multimodal approach. EPJ Data Sci 12(1):55. https://doi.org/10.1140/epjds/s13688-023-00429-y

Girvan M, Newman ME (2002) Community structure in social and biological networks. Proc Natl Acad Sci 99(12):7821–7826. https://doi.org/10.1073/pnas.122653799

Guimera R, Danon L, Diaz-Guilera A, Giralt F, Arenas A (2003) Self-similar community structure in a network of human interactions. Phys Rev E 68(6):065103. https://doi.org/10.1103/PhysRevE.68.065103

Gupta S, Manchanda S, Bedathur S, Ranu S (2022) Tigger: scalable generative modelling for temporal interaction graphs. In: Proceedings of the AAAI conference on artificial intelligence, vol 36, pp 6819–6828. https://doi.org/10.1609/aaai.v36i6.20638

Hoffman M, Block P, Elmer T, Stadtfeld C (2020) A model for the dynamics of face-to-face interactions in social groups. Netw Sci 8(S1):4–25. https://doi.org/10.1017/nws.2020.3

Holme P (2013) Epidemiologically optimal static networks from temporal network data. PLoS Comput Biol 9(7):1003142. https://doi.org/10.1371/journal.pcbi.1003142

Iacopini I, Petri G, Baronchelli A, Barrat A (2022) Group interactions modulate critical mass dynamics in social convention. Commun Phys 5(1):64. https://doi.org/10.1038/s42005-022-00845-y

Isella L, Romano M, Barrat A, Cattuto C, Colizza V, Broeck W, Gesualdo F, Pandolfi E, Ravà L, Rizzo C, Tozzi AE (2011) Close encounters in a pediatric ward: measuring face-to-face proximity and mixing patterns with wearable sensors. PLoS ONE 6(2):e17144. https://doi.org/10.1371/journal.pone.0017144

Isella L, Stehlé J, Barrat A, Cattuto C, Pinton J-F, Broeck W (2011) What's in a crowd? Analysis of face-to-face behavioral networks. J Theor Biol 271(1):166–180. https://doi.org/10.1016/j.jtbi.2010.11.033

Jackson MO (2008) Social and economic networks. Princeton University Press, Princeton. https://doi.org/10.2307/j.ctvcm4gh1

Jolliffe IT, Cadima J (2016) Principal component analysis: a review and recent developments. Philos Trans R Soc Math Phys Eng Sci 374(2065):20150202. https://doi.org/10.1098/rsta.2015.0202

Karrer B, Newman ME (2009) Random graph models for directed acyclic networks. Phys Rev E-Stat Nonlinear Soft Matter Phys 80(4):046110. https://doi.org/10.1103/PhysRevE.80.046110

Kovács IA, Luck K, Spirohn K, Wang Y, Pollis C, Schlabach S, Bian W, Kim D-K, Kishore N, Hao T et al (2019) Network-based prediction of protein interactions. Nat Commun 10(1):1240. https://doi.org/10.1038/s41467-019-09177-y

Laurent G, Saramäki J, Karsai M (2015) From calls to communities: a model for time-varying social networks. Eur Phys J B 88:1–10. https://doi.org/10.1140/epjb/e2015-60481-x

Lazer D, Pentland A, Adamic L, Aral S, Barabási A-L, Brewer D, Christakis N, Contractor N, Fowler J, Gutmann M et al (2009) Computational social science. Science 323(5915):721–723. https://doi.org/10.1126/science.1167742

Leskovec J, Lang KJ, Dasgupta A, Mahoney MW (2008) Statistical properties of community structure in large social and information networks. In: Proceedings of the 17th international conference on World Wide Web, pp 695–704. https://doi.org/10.1145/1367497.1367591

Limnios S, Elliott A, Cucuringu M, Reinert G (2022) Random walk based conditional generative model for temporal networks with attributes

Liu P, Sariyüce AE (2023) Using motif transitions for temporal graph generation. In: Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining, pp 1501–1511.https://doi.org/10.1145/3580305.3599540

Longa A, Cencetti G, Lepri B, Passerini A (2022) An efficient procedure for mining egocentric temporal motifs. Data Min Knowl Disc 36(1):355–378. https://doi.org/10.1007/s10618-021-00803-2

Longa A, Cencetti G, Lehmann S, Passerini A, Lepri B (2024) Generating fine-grained surrogate temporal networks. Commun Phys 7(1):22. https://doi.org/10.1038/s42005-023-01517-1
Lotito QF, Montresor A, Battiston F (2024) Multiplex measures for higher-order networks. Appl Netw Sci 9(1):55. https://doi.org/10.1007/s41109-024-00665-9
Mastrandrea R, Fournet J, Barrat A (2015) Contact patterns in a high school: a comparison between data collected using wearable sensors, contact diaries and friendship surveys. PLoS ONE 10(9):0136497. https://doi.org/10.1371/journal.pone.0136497
Mauro G, Luca M, Longa A, Lepri B, Pappalardo L (2022) Generating mobility networks with generative adversarial networks. EPJ Data Sci 11(1):58. https://doi.org/10.1140/epjds/s13688-022-00372-4
Nematzadeh A, Ferrara E, Flammini A, Ahn Y-Y (2014) Optimal network modularity for information diffusion. Phys Rev Lett 113(8):088701. https://doi.org/10.1103/PhysRevLett.113.088701
Nepal S, Wang W, Vojdanovski V, Huckins JF, daSilva A, Meyer M, Campbell A (2022) COVID student study: A year in the life of college students during the COVID-19 pandemic through the lens of mobile phone sensing. In: Proceedings of the 2022 CHI conference on human factors in computing systems, pp 1–19. https://doi.org/10.1145/3491102.3502043
Newman M (2010) Networks: an introduction. Oxford University Press, Oxford. https://doi.org/10.1093/acprof:oso/9780199206650.001.0001
Ozella L, Gauvin L, Carenzo L, Quaggiotto M, Ingrassia PL, Tizzoni M, Panisson A, Colombo D, Sapienza A, Kalimeri K et al (2019) Wearable proximity sensors for monitoring a mass casualty incident exercise: feasibility study. J Med Internet Res 21(4):12251. https://doi.org/10.2196/12251
Peixoto TP, Rosvall M (2017) Modelling sequences and temporal networks with dynamic community structures. Nat Commun 8(1):582. https://doi.org/10.1038/s41467-017-00148-9
Peng H, Nematzadeh A, Romero DM, Ferrara E (2020) Network modularity controls the speed of information diffusion. Phys Rev E 102(5):052316. https://doi.org/10.1103/PhysRevE.102.052316
Peng Y, Zhao Y, Hu J (2023) On the role of community structure in evolution of opinion formation: a new bounded confidence opinion dynamics. Inf Sci 621:672–690. https://doi.org/10.1016/j.ins.2022.11.101
Pereira LR, Lopes RJ, Louçã J (2020) Syntgen: a system to generate temporal networks with user-specified topology. J Complex Netw 8(4):039. https://doi.org/10.1093/comnet/cnz039
Perra N, Gonçalves B, Pastor-Satorras R, Vespignani A (2012) Activity driven modeling of time varying networks. Sci Rep 2(1):469. https://doi.org/10.1038/srep00469
Purohit S, Holder LB, Chin G (2018) Temporal graph generation based on a distribution of temporal motifs. In: Proceedings of the 14th international workshop on mining and learning with graphs, vol 7
Rajeh S, Yassin A, Jaber A, Cherifi H (2021) Analyzing community-aware centrality measures using the linear threshold model. In: International conference on complex networks and their applications. Springer, pp 342–353. https://doi.org/10.1007/978-3-030-93409-5_29
Salathé M, Jones JH (2010) Dynamics and control of diseases in networks with community structure. PLoS Comput Biol 6(4):1000736. https://doi.org/10.1371/journal.pcbi.1000736
Sekara V, Stopczynski A, Lehmann S (2016) Fundamental structures of dynamic social networks. Proc Natl Acad Sci 113(36):9977–9982. https://doi.org/10.1073/pnas.1602803113
Starnini M, Lepri B, Baronchelli A, Barrat A, Cattuto C, Pastor-Satorras R (2017) Robust modeling of human contact networks across different scales and proximity-sensing techniques. In: Social informatics: 9th international conference, SocInfo 2017, Oxford, UK, September 13–15, 2017, Proceedings, Part I 9, pp 536–551. https://doi.org/10.1007/978-3-319-67217-5_32. Springer
Stehlé J, Voirin N, Barrat A, Cattuto C, Isella L, Pinton J-F, Quaggiotto M, Broeck W, Régis C, Lina B et al (2011) High-resolution measurements of face-to-face contact patterns in a primary school. PLoS ONE 6(8):23176. https://doi.org/10.1371/journal.pone.0023176
Stopczynski A, Sekara V, Sapiezynski P, Cuttone A, Madsen MM, Larsen JE, Lehmann S (2014) Measuring large-scale social networks with high resolution. PLoS ONE 9(4):95978. https://doi.org/10.1371/journal.pone.0095978
Stopczynski A, Sapiezynski P, Pentland A, Lehmann S (2015) Temporal fidelity in dynamic social networks. Eur Phys J B 88:1–6. https://doi.org/10.1140/epjb/e2015-60549-7
Traag VA, Waltman L, Van Eck NJ (2019) From Louvain to Leiden: guaranteeing well-connected communities. Sci Rep 9(1):1–12. https://doi.org/10.1038/s41598-019-41695-z
Vanhems P, Barrat A, Cattuto C, Pinton J-F, Khanafer N, Régis C, Kim B-A, Comte B, Voirin N (2013) Estimating potential infection transmission routes in hospital wards using wearable proximity sensors. PLoS ONE 8(9):73970. https://doi.org/10.1371/journal.pone.0073970
Vico Fallani F, Richiardi J, Chavez M, Achard S (2014) Graph analysis of functional brain networks: practical issues in translational neuroscience. Philos Trans R Soc B Biol Sci 369(1653):20130521. https://doi.org/10.1098/rstb.2013.0521
Zeno G, Neville J (2023) Dyane: dynamic attributed node roles generative model. In: Proceedings of the 32nd ACM international conference on information and knowledge management, pp 3144–3153. https://doi.org/10.1145/3583780.3614858
Zeno G, La Fond T, Neville J (2021) Dymond: dynamic motif-nodes network generative model. In: Proceedings of the web conference 2021, pp 718–729. https://doi.org/10.1145/3442381.3450102
Zhang L, Zhao L, Qin S, Pfoser D, Ling C (2021) Tg-gan: continuous-time temporal graph deep generative models with time-validity constraints. In: Proceedings of the web conference 2021, pp 2104–2116. https://doi.org/10.1145/3442381.3449818
Zhou D, Zheng L, Han J, He J (2020) A data-driven graph generative model for temporal interaction networks. In: Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, pp 401–411. https://doi.org/10.1145/3394486.3403082

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